pfi

Python for Fixed Income

COURSE 1 · PYTHON FOUNDATIONS

More Visualization

WEEK 3, DAY 2

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The desk lead's note: three asks before the meeting

Your seven charts went round the desk. Three asks before the meeting. Put a line through a scatter and give me its slope, in units. Show me how many bonds stand behind each shape. Then make the charts meeting-ready: our colours, numbers I can read, the bond everyone asks about marked, and a file for the slide. No chart says more than the data.

The issuers are invented; a line fitted here summarizes this file and forecasts nothing.

12.1 Yesterday's Book in One Cell

```
sns.set_theme(style='whitegrid', palette=['#2E7D5B', '#9AA5B1', '#1E4D3A', '#52606D'],
              rc={'boxplot.medianprops.color': '#1E4D3A'})
holdings = pd.read_csv(data_folder + 'holdings.csv')
ratings = pd.read_csv(data_folder + 'ratings.csv')
book = pd.merge(holdings, ratings, on='rating')
bucket_order = ratings.sort_values('score')['bucket'].unique().tolist()
print(book.shape)
(200, 22)
```

Three asks, in order: a slope, the bonds, a chart fit to leave the desk.

Same table, same credit order, same greens: today adds a slope, the bonds, and the finish.

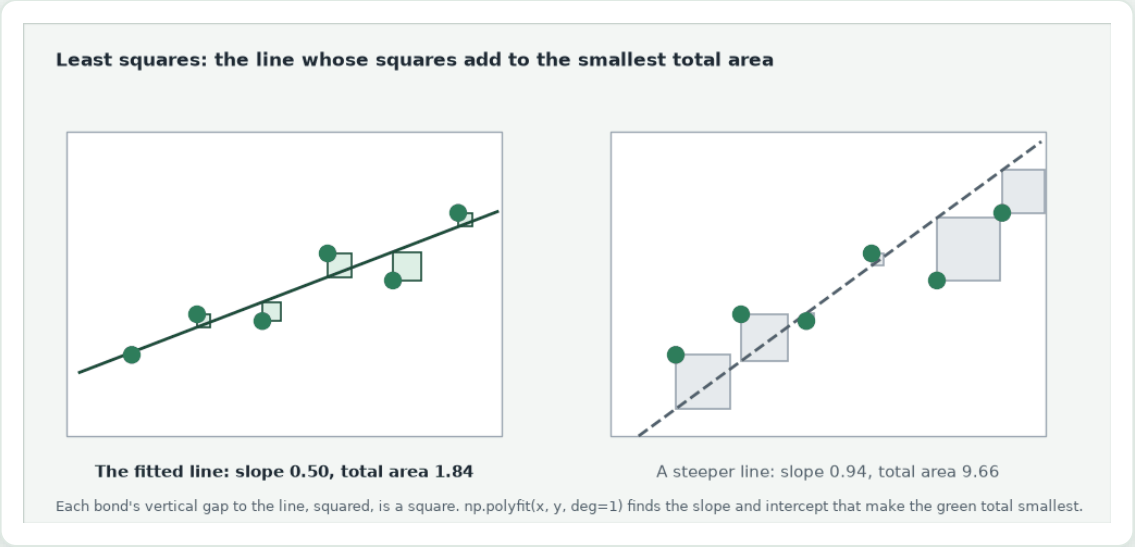
12.2 A line through coupon and yield, in units

A regression line is "y = intercept + slope * x". The slope is the change in y per unit of x, in y's unit per x's unit; the intercept is the line's height where x is zero.

```
slope, intercept = np.polyfit(book['coupon'], book['yield_pct'], deg=1)
print('Slope:', round(slope, 2), 'points of yield per point of coupon')
print('Intercept:', round(intercept, 2), 'percent')
print('The line at 5 and 8 percent coupons:',
      round(intercept + slope * 5, 2), round(intercept + slope * 8, 2))
print('Lowest coupon in the book:', book['coupon'].min())
Slope: 0.88 points of yield per point of coupon
Intercept: 0.85 percent
The line at 5 and 8 percent coupons: 5.26 7.91
Lowest coupon in the book: 3.25
```

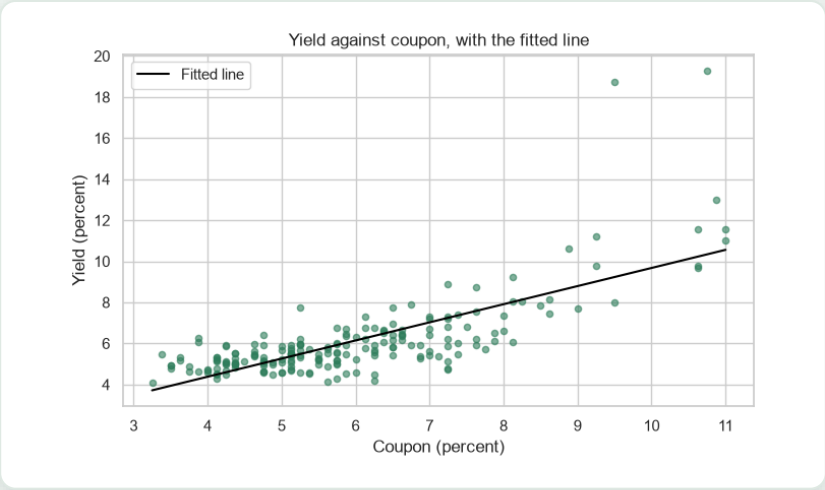
```
slope, intercept = np.polyfit(book['coupon'], book['yield_pct'])
TypeError: polyfit() missing 1 required positional argument: 'deg'
```

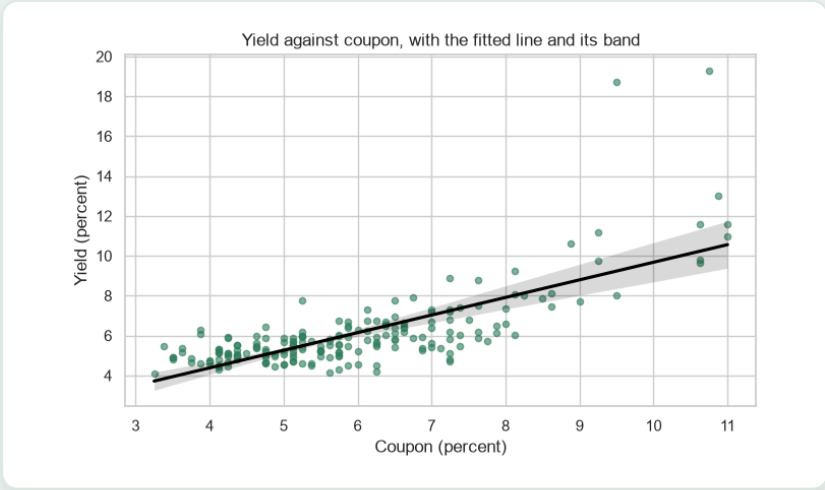
A "TypeError", read from the bottom as in week 1's Day 3: NumPy will not guess the degree.



One more point of coupon goes with 0.88 more of yield, so 5.26 and 7.91 sit 3 x 0.88 apart: points on the line, not any bond's yield. The intercept, 0.85, is the height at a zero coupon, far left of the lowest bond's 3.25.

The line over the bonds, its band, and R squared





Two bonds above 18 percent sit far over the line; squaring each gap gives a far bond a large say.

The band, built like Visualization's from 1,000 random redraws, is narrow where the bonds crowd and wide at high coupons; it is about the line, not the bonds.

```
print('R squared:', round(book['coupon'].corr(book['yield_pct']) ** 2, 2))
R squared: 0.55
```

R squared, from 0 to 1, is the share of the variation in y that a straight least-squares line on x explains: 55 percent of yield's here.

Excel side by side. Coupon is column G and yield column L.

Job	Python	Excel
Slope	`np.polyfit(x, y, deg=1)[0], 0.88`	=SLOPE(L2:L201, G2:G201)
Intercept	`np.polyfit(x, y, deg=1)[1], 0.85`	=INTERCEPT(L2:L201, G2:G201)
R squared	`x.corr(y) ** 2, 0.55`	=RSQ(L2:L201, G2:G201)
The line on the chart	`sns.regplot()`	Chart Elements, Trendline, Linear

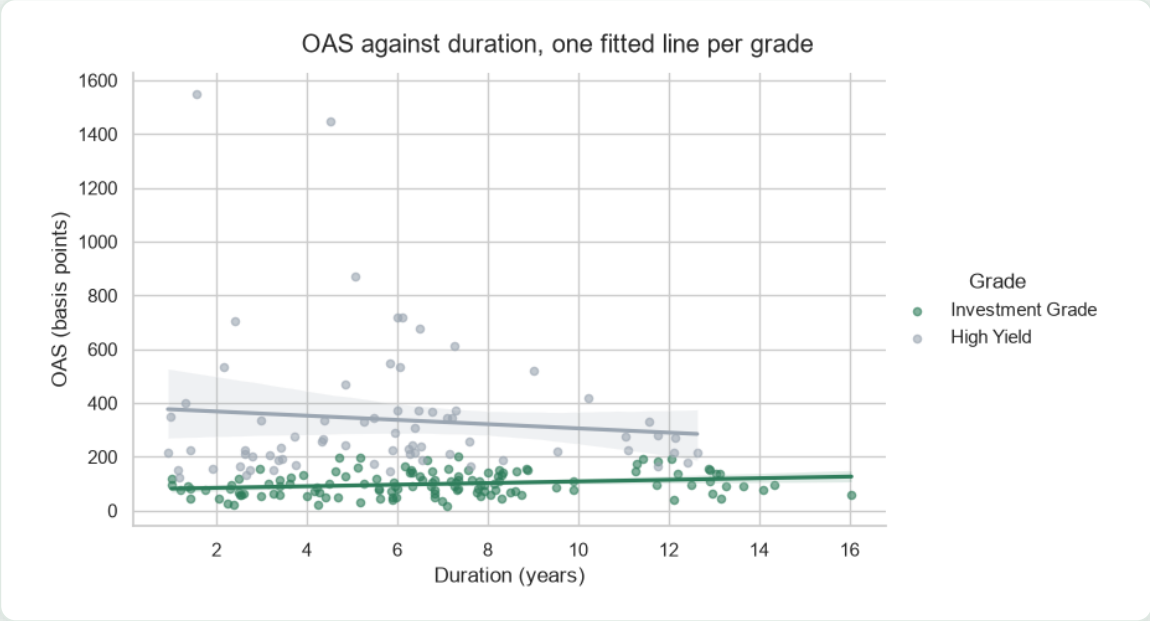
Excel takes y first and NumPy x first; Excel's trendline draws no band.

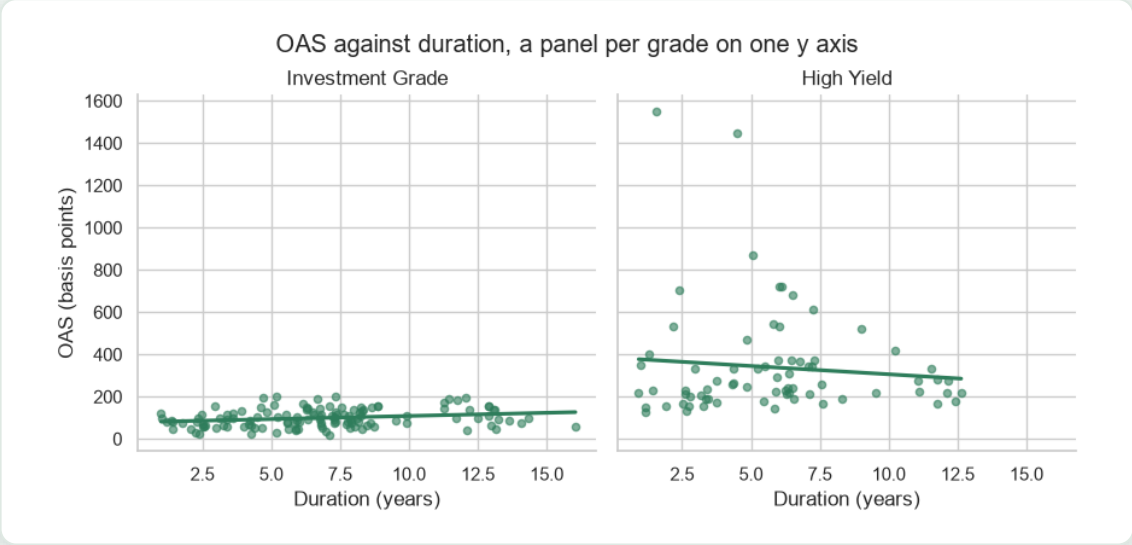
Slope, intercept, R squared, each with its unit: print them beside any line.

One line per grade, fitted on its own bonds

```
for grade in ['Investment Grade', 'High Yield']:
    part = book[book['grade'] == grade]
    print(grade, round(np.polyfit(part['duration'], part['oas'], deg=1)[0], 2), 'bp per year')
Investment Grade 2.99 bp per year
High Yield -7.86 bp per year
```

Q. The Visualization notebook split this cloud by grade. Fit each grade on its own bonds.

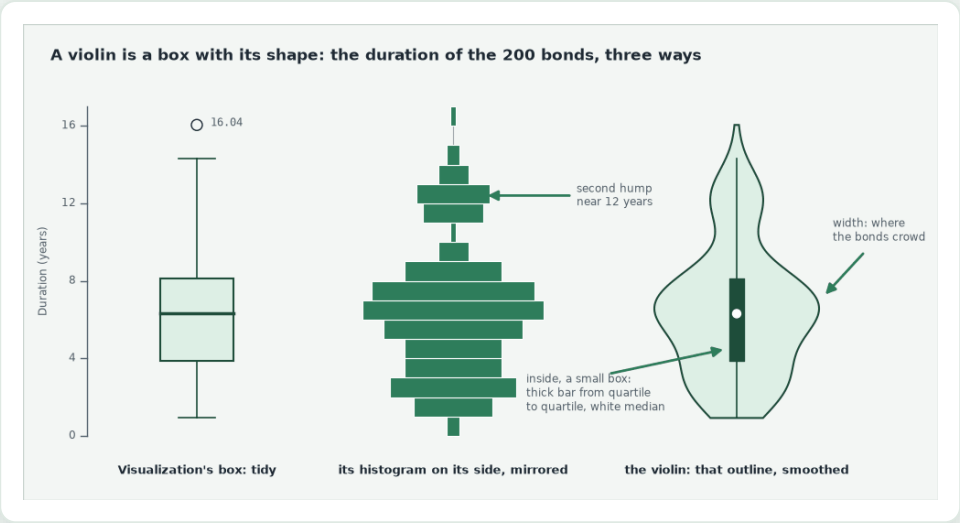




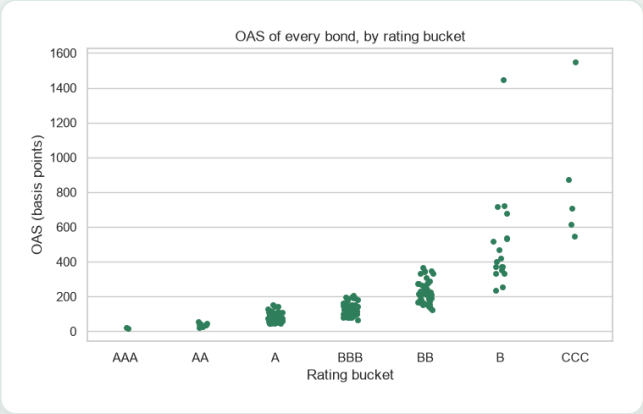
The High Yield band is wider: fewer bonds, spread further apart.

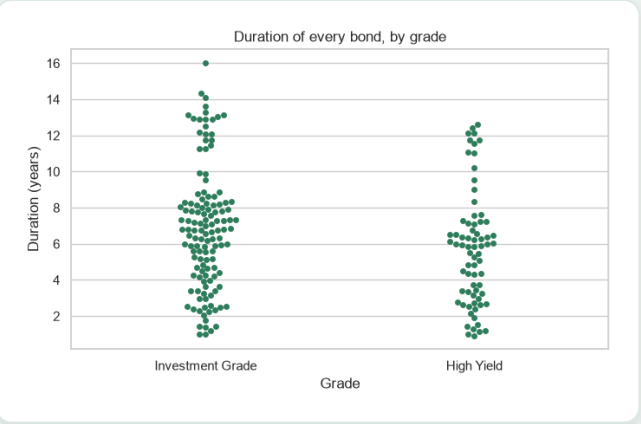
12.3 A violin is a box with its shape; dots are the bonds

A violin plot is a box with its shape: a smoothed histogram, mirrored, wide where the bonds crowd.



Dots show how many bonds stand behind a shape, and one function draws all of these.





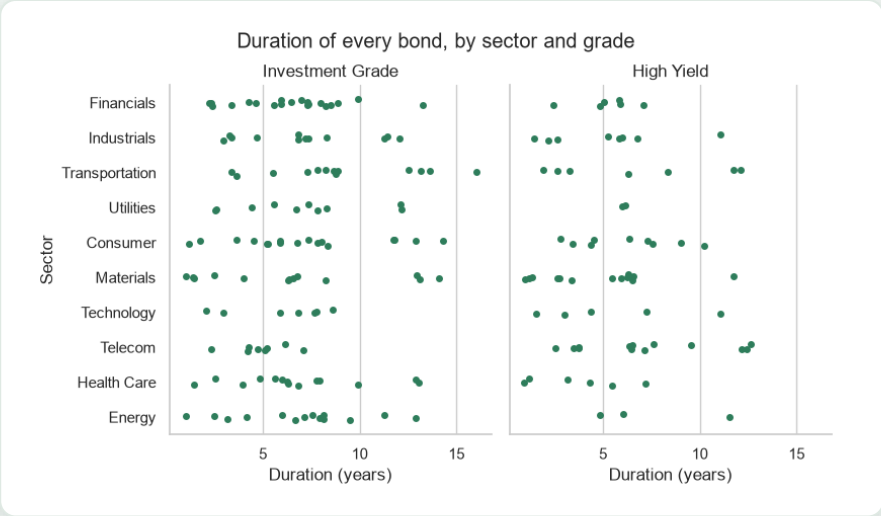
The dots count for you: 2 at AAA and 5 at CCC, against 56, 62 and 49 in the middle, where Visualization's boxes drew AAA's 2 like BBB's 62. The sideways nudge, jitter, means nothing; only height is data.

A swarm is a violin's outline built of bonds: a bulge near 12 years, 16.04 alone on top.

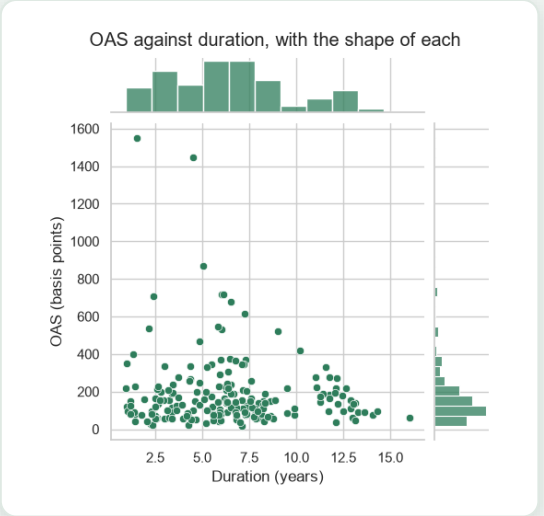
Excel side by side. OAS is column M and grade column V.

Job	Python	Excel
Shape of each group	`sns.violinplot()`	no violin chart; Box and Whisker is nearest
Every bond as a dot	`sns.stripplot()`	Box and Whisker, Format Data Series, Show inner points
One group's lowest	19	=MINIFS(M:M, V:V, "Investment Grade"); MAXIFS gives 204

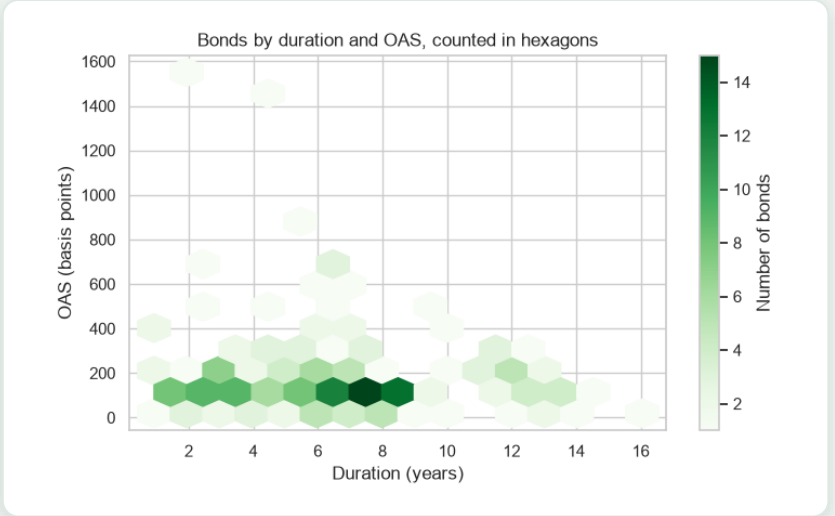
12.4 and 12.5 Panels, and a joint plot



Read a row across: Utilities has 10 Investment Grade dots and 2 High Yield. The dot furthest right, in Transportation, is the 16.04-year bond.



Along the top is duration's histogram; down the right, Visualization's OAS histogram on its side, 164 below 250 bp. The darkest hexagon sits low, near 7 to 8 years; the widest bonds are pale singles far above.



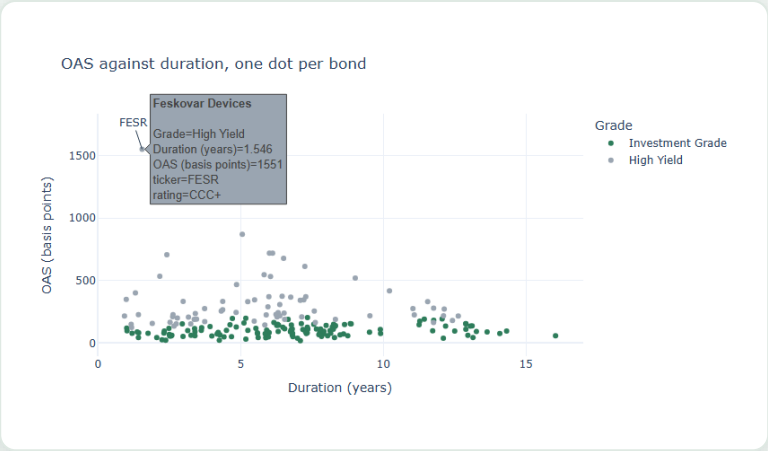
Excel side by side

A panel per group is one chart per group, each from the filtered table; =COUNTIFS(E:E, "Utilities", V:V, "High Yield") gives the 2.

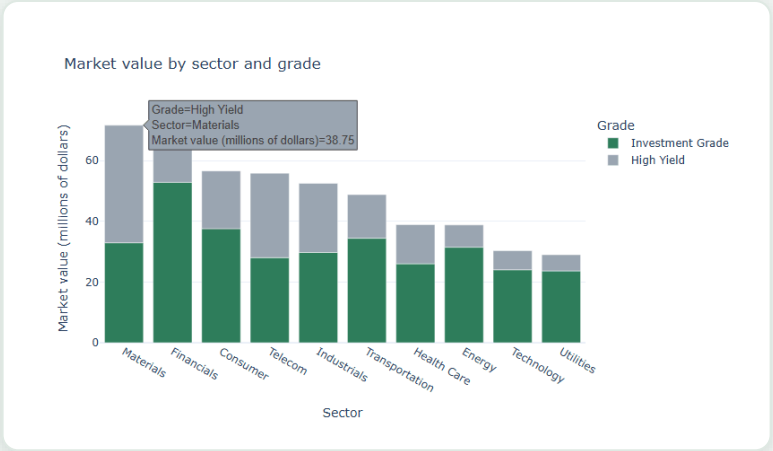
Excel side by side

No single chart: a Scatter with two Histograms beside it, and =COUNTIF(M:M, "<250") for the 164.

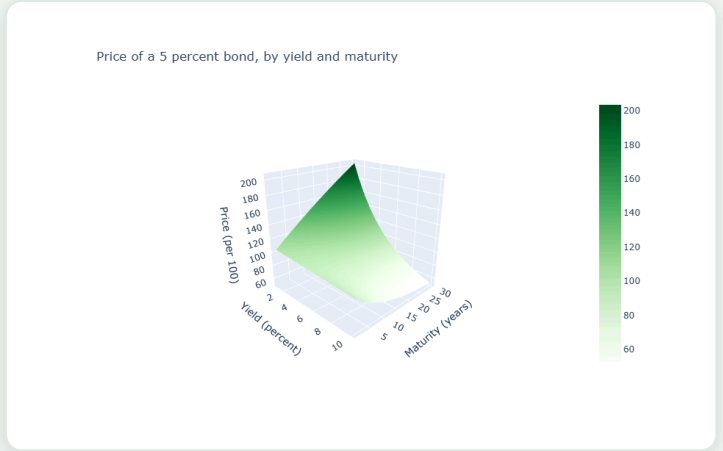
12.6 Charts that answer the mouse, and a surface



The top dot is Feskovar Devices, an invented issuer: FESR, rated CCC+, at 1.546 years and 1,551 bp.



Materials leads with 71.70 million, then Financials with 68.22, as in Visualization; point at a piece for its value.



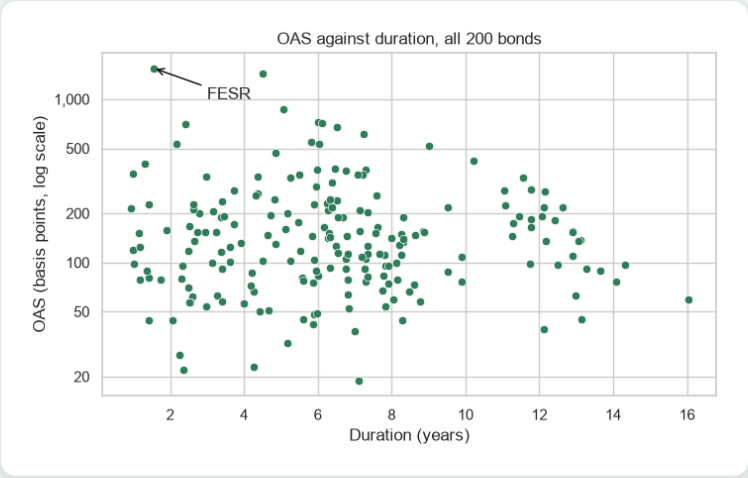
The prices lie nearly flat along the 1-year edge, 95.35 to 103.97, fan out at 30 years, and stay at 100 along the 5 percent line.

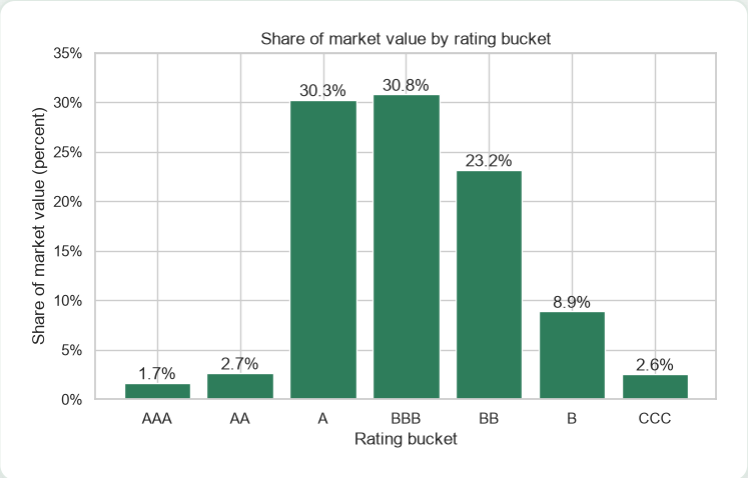
Excel side by side. Market value is column S, sector E, grade V.

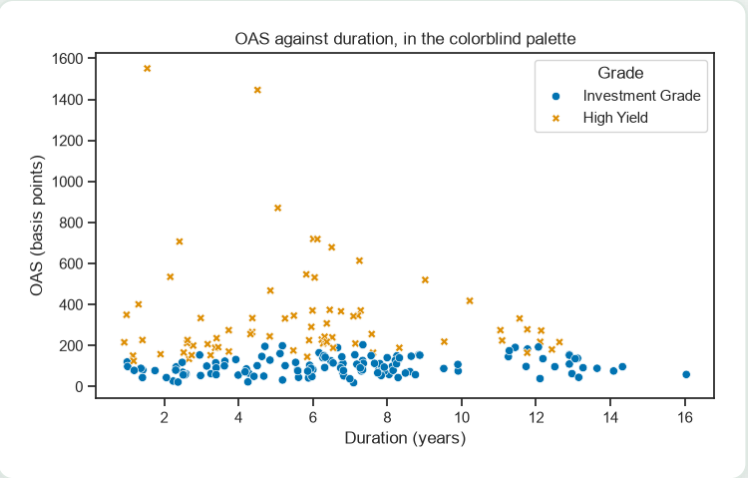
Job	Python	Excel
Read a point	`the hover box`	rest the pointer: a tip with the series and value
One stacked piece	`Consumer, High Yield, 19.02`	=SUMIFS(S:S, E:E, "Consumer", V:V, "High Yield") / 1000000
A bond's price	`bond_price(10, 30), 52.68`	=PRICE(DATE(2026,9,15), DATE(2056,9,15), 0.05, 0.1, 100, 2)

On screen a chart can answer the mouse; a meeting still wants a still picture in the desk's look.

12.7 A log axis, shares, and colours for all







All 200 are on, and 50 to 100 stands as tall as 500 to 1,000: each is a doubling. OAS runs from 19 to 1,551 bp, the kind of range a log axis is for.

The "ticks" look, with no grid, lasts for this chart alone.

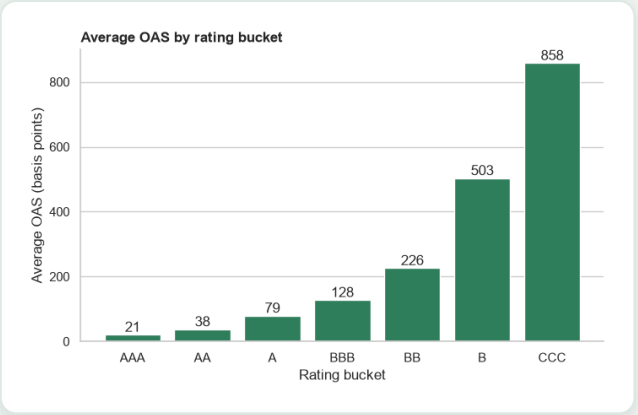
The Desk's Colours

```
grade_colours = {'Investment Grade': '#2E7D5B', 'High Yield': '#52606D'}
```

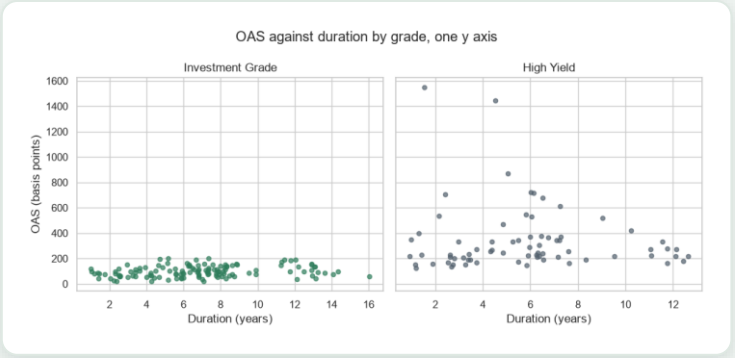
```
ax.bar_label(bars, fmt='%.1f%%')
ax.yaxis.set_major_formatter(PercentFormatter(xmax=100, decimals=0))
```

One desk style, one scale, saved for the slide

```
def desk_style(ax, title, xlabel, ylabel):
    ax.set_title(title, loc='left', fontweight='bold')
    ax.set_xlabel(xlabel)
    ax.set_ylabel(ylabel)
    ax.spines['top'].set_visible(False)
    ax.spines['right'].set_visible(False)
    ax.grid(axis='x', visible=False)
    ax.set_axisbelow(True)
```



These are Visualization's bars, 21 bp at AAA to 858 at CCC, in the desk's look: one call styles any chart, and one edit restyles the pack.



On one scale Investment Grade is a flat band under 204 bp; without "sharey=True" both panels would look equally tall.

Excel side by side

Job	Python	Excel
Axis limits	`ax.set_ylim(0, 600)`	Format Axis, Bounds, Minimum and Maximum
Thousands separators	`ax.yaxis.set_major_formatter('{x:,.0f}')`	Format Axis, Number, Format Code #,##0
A value on each bar	`ax.bar_label(bars)`	Chart Elements, Data Labels
Save as a picture	`fig.savefig('name.png')`	right-click the chart, Save as Picture

One function holds the look; one line writes the file.

Predict first: a slope through a cloud, a violin's end

Predict 1

The Visualization notebook put duration and OAS at -0.11. What slope does "np.polyfit(book['duration'], book['oas'], deg=1)" give? A) Positive: longer bonds carry more spread. B) Negative, and small against OAS's range of 19 to 1,551 bp. C) Exactly zero.

```
duration_slope, duration_intercept = np.polyfit(book['duration'], book['oas'], deg=1)
print('Slope:', round(duration_slope, 2), 'bp per year  R squared:',
      round(book['duration'].corr(book['oas']) ** 2, 2))
fig, ax = plt.subplots(figsize=(8, 4.5))
sns.regplot(data=book, x='duration', y='oas', seed=0,
            scatter_kws={'s': 20, 'alpha': 0.6}, line_kws={'color': 'black'}, ax=ax)
ax.set_title('OAS against duration, with the fitted line')
ax.set_xlabel('Duration (years)')
ax.set_ylabel('OAS (basis points)')
plt.show()
```

The slope is the change in y per unit of x, in y's unit per x's unit; the intercept is the line's height where x is zero.

Predict 2

No bond's OAS is below 19 bp. Where does the bottom of the Investment Grade violin end in "sns.violinplot(data=book, x='grade', y='oas')"? A) At 19 bp, the lowest bond. B) At 0. C) Below 0.

```
fig, ax = plt.subplots(figsize=(8, 4.5))
sns.violinplot(data=book, x='grade', y='oas', ax=ax)
ax.axhline(0, color='black', linestyle='--')
ax.set_title('OAS by grade')
ax.set_xlabel('Grade')
ax.set_ylabel('OAS (basis points)')
plt.show()
```

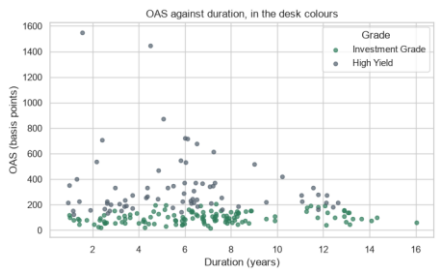
A violin plot is a box with its shape: a smoothed histogram, mirrored, wide where the bonds crowd.

Predict first: what a narrower axis does to 8 bonds

Predict 3

8 bonds have an OAS above 600 bp. What happens to them after "ax.set_ylim(0, 600)"? A) They are deleted from "book". B) They stay in "book", are not drawn, and nothing on the chart says so. C) An error: data falls outside the axis.

```
fig, ax = plt.subplots(figsize=(8, 4.5))
sns.scatterplot(data=book, x='duration', y='oas', hue='grade', palette=grade_colours, ax=ax)
ax.set_ylim(0, 600)
ax.set_xlim(0, 17)
ax.legend(title='Grade', loc='upper right')
ax.set_title('OAS against duration (8 bonds above 600 bp not shown)')
ax.set_xlabel('Duration (years)')
ax.set_ylabel('OAS (basis points)')
plt.show()
print('Rows in book:', len(book), '  OAS above 600 bp:', (book['oas'] > 600).sum())
```



The two widest bonds stretch the axis to 1,551 and press the crowd flat.

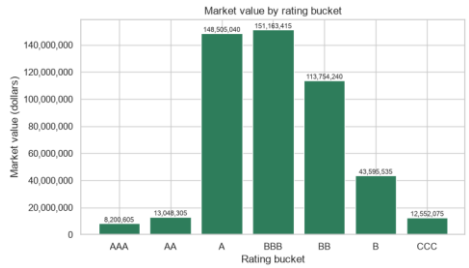
Q. Give each grade one colour for every chart in the pack.

Predict first: two bars on an axis from 147

Predict 4

A holds 148.51 million and BBB 151.16 million. With the value axis starting at 147 million, how many times as tall as A's bar is BBB's? Write down a number.

```
two = value_by_bucket[['A', 'BBB']] / 1_000_000
fig, axes = plt.subplots(1, 2, figsize=(9, 4.5))
for ax, title in zip(axes, ['Value axis from zero', 'Value axis from 147: misleading']):
    bars = ax.bar(two.index, two.values)
    ax.bar_label(bars, fmt='%.2f')
    ax.set_title(title)
axes[1].set_ylim(147, 152)
axes[0].set_ylabel('Market value (millions of dollars)')
plt.tight_layout()
plt.show()
```



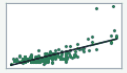
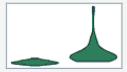
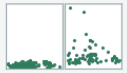
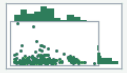

```
ax.bar_label(bars, fmt='{:,.0f}', fontsize=8)
ax.yaxis.set_major_formatter('{x:,.0f}')
```

The note, answered

Asked	Chart	Answer
A slope, in units	regression plot	0.88 points of yield per point of coupon, R squared 0.55
The bonds behind the shapes	strip plot	2 bonds at AAA against 62 at BBB

Every chart today trusted the file; the next notebook opens one that should not be trusted and uses these charts to find what is wrong.

Today's charts, and the question each one answers

The question	Today's chart	Its name	Python	Excel
How much y moves with x		**Regression plot**	`np.polyfit` `sns.regplot, sns.lmplot`	Trendline, SLOPE
Each group's shape		**Violin plot**	`sns.violinplot`	Box and Whisker, nearest
Every bond in each group		**Strip, swarm**	`sns.stripplot` `sns.swarmplot`	Show inner points
One chart per group		**Panels**	`sns.catplot(col=)` `plt.subplots(1, 2)`	a chart per group
Two columns, and each one's shape		**Joint plot**	`sns.jointplot` `ax.hexbin`	a Scatter, two Histograms
Which bond is that dot?		**Interactive**	`px.scatter, px.box` `px.bar`	the hover tip
A height from two inputs		**3D, surface**	`px.scatter_3d` `go.Surface`	3-D Surface

The habits of a chart that leaves the desk:

One unit per chart; back colour with a shape.

The Visualization notebook's picture holds the first seven charts; today added these.